

ENPM 667 Final Project

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1 Obtain the equations of motion for the system and the corresponding nonlinear state-space representation.

To calculate the equations of motion we need to calculate the the Euler-Lagrange equations using $\mathcal{L} = K - P$, where K is Kinetic Energy and P is Potential Energy.

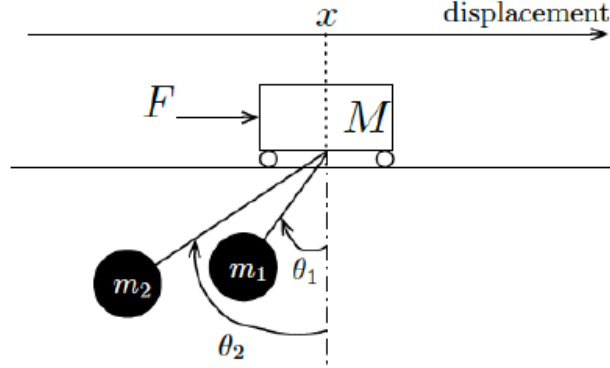


Figure 1: Cart with 2 suspended weights

1.1 Calculation of Kinetic energy

There are 3 mass objects defined by M, m_1 and m_2 . Total Kinetic Energy will be: $K = K_1 + K_2 + K_3$, where K_1, K_2 , and K_3 are the kinetic energies of mass M, m_1 , and m_2 , respectively. Lets calculate each of these kinetic energies first.

Cart Mass (M):

$$K_1 = \frac{1}{2} M \dot{x}^2$$

Mass m_1 : The position of m_1 is defined as:

$$\begin{aligned} x_1 &= x - l_1 \sin \theta_1 \\ y_1 &= -l_1 \cos \theta_1 \end{aligned}$$

Taking the derivatives:

$$\begin{aligned}\dot{x}_1 &= \dot{x} - l_1 \dot{\theta}_1 \cos \theta_1 \\ \dot{y}_1 &= l_1 \dot{\theta}_1 \sin \theta_1\end{aligned}$$

The kinetic energy K_2 is:

$$\begin{aligned}K_2 &= \frac{1}{2}m_1(\dot{x}_1^2 + \dot{y}_1^2) \\ &= \frac{1}{2}m_1[(\dot{x} - l_1 \dot{\theta}_1 \cos \theta_1)^2 + (l_1 \dot{\theta}_1 \sin \theta_1)^2] \\ &= \frac{1}{2}m_1[\dot{x}^2 - 2\dot{x}l_1 \dot{\theta}_1 \cos \theta_1 + l_1^2 \dot{\theta}_1^2 (\cos^2 \theta_1 + \sin^2 \theta_1)] \\ &= \frac{1}{2}m_1(\dot{x}^2 - 2\dot{x}l_1 \dot{\theta}_1 \cos \theta_1 + l_1^2 \dot{\theta}_1^2)\end{aligned}$$

Mass m_2 : Similarly, for mass m_2 :

$$K_3 = \frac{1}{2}m_2(\dot{x}^2 - 2\dot{x}l_2 \dot{\theta}_2 \cos \theta_2 + l_2^2 \dot{\theta}_2^2)$$

Total Kinetic Energy: Summing these components and rearranging:

$$K = \frac{1}{2}(M + m_1 + m_2)\dot{x}^2 - \dot{x}(m_1 l_1 \dot{\theta}_1 \cos \theta_1 + m_2 l_2 \dot{\theta}_2 \cos \theta_2) + \frac{1}{2}m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2}m_2 l_2^2 \dot{\theta}_2^2 \quad (1)$$

1.2 Potential Energy Calculation

The potential energy P is calculated as:

$$\begin{aligned}P_1(\text{mass } M) &= 0 \\ P_2(\text{mass } m_1) &= -m_1 g l_1 \cos \theta_1 \\ P_3(\text{mass } m_2) &= -m_2 g l_2 \cos \theta_2\end{aligned}$$

Total Potential Energy:

$$P = -m_1 g l_1 \cos \theta_1 - m_2 g l_2 \cos \theta_2 \quad (2)$$

1.3 Lagrangian Equation

Using $\mathcal{L} = K - P$ and substituting equation (1) and the potential energy:

$$\mathcal{L} = \text{Eq (3)} - \text{Eq (4)}$$

Expanded form:

$$\begin{aligned}\mathcal{L} &= \frac{1}{2}(m_1 + m_2 + M)\dot{x}^2 - \dot{x}(m_1 l_1 \dot{\theta}_1 \cos \theta_1 + m_2 l_2 \dot{\theta}_2 \cos \theta_2) \\ &\quad + \frac{1}{2}m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2}m_2 l_2^2 \dot{\theta}_2^2 + m_1 g l_1 \cos \theta_1 + m_2 g l_2 \cos \theta_2\end{aligned} \quad (3)$$

1.4 Euler-Lagrange Equations of Motion

We solve the Euler-Lagrange equations for x , θ_1 , and θ_2 .

Equation for x

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} = F \quad (4)$$

Derivatives:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= 0 \\ \frac{\partial \mathcal{L}}{\partial \dot{x}} &= (m_1 + m_2 + M)\dot{x} - (m_1 l_1 \dot{\theta}_1 \cos \theta_1 + m_2 l_2 \dot{\theta}_2 \cos \theta_2) \end{aligned}$$

Taking the time derivative of the above equation we get $\frac{d}{dt}$:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = (m_1 + m_2 + M)\ddot{x} - [m_1 l_1 \ddot{\theta}_1 \cos \theta_1 - m_1 l_1 \dot{\theta}_1^2 \sin \theta_1 + m_2 l_2 \ddot{\theta}_2 \cos \theta_2 - m_2 l_2 \dot{\theta}_2^2 \sin \theta_2]$$

Now Substituting these derivatives in equation (4) we get:

$$(m_1 + m_2 + M)\ddot{x} - [m_1 l_1 \ddot{\theta}_1 \cos \theta_1 - m_1 l_1 \dot{\theta}_1^2 \sin \theta_1 + m_2 l_2 \ddot{\theta}_2 \cos \theta_2 - m_2 l_2 \dot{\theta}_2^2 \sin \theta_2] = F$$

On rearranging the above equation we get:

$$(m_1 + m_2 + M)\ddot{x} - m_1 l_1 (\ddot{\theta}_1 \cos \theta_1 - \dot{\theta}_1^2 \sin \theta_1) - m_2 l_2 (\ddot{\theta}_2 \cos \theta_2 - \dot{\theta}_2^2 \sin \theta_2) = F \quad (5)$$

Equation for θ_1

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1} \right) - \frac{\partial \mathcal{L}}{\partial \theta_1} = 0 \quad (6)$$

Derivatives:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \theta_1} &= -\dot{x} m_1 l_1 \dot{\theta}_1 (-\sin \theta_1) - m_1 g l_1 \sin \theta_1 \\ &= m_1 l_1 \dot{x} \dot{\theta}_1 \sin \theta_1 - m_1 g l_1 \sin \theta_1 \\ \frac{\partial \mathcal{L}}{\partial \dot{\theta}_1} &= -\dot{x} m_1 l_1 \cos \theta_1 + m_1 l_1^2 \dot{\theta}_1 \end{aligned}$$

Time derivative:

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_1} \right) &= -[\ddot{x} m_1 l_1 \cos \theta_1 + \dot{x} m_1 l_1 (-\sin \theta_1) \dot{\theta}_1] + m_1 l_1^2 \ddot{\theta}_1 \\ &= -m_1 l_1 \ddot{x} \cos \theta_1 + m_1 l_1 \dot{x} \dot{\theta}_1 \sin \theta_1 + m_1 l_1^2 \ddot{\theta}_1 \end{aligned}$$

Now Substituting these derivatives in equation (6) we get:

$$(-m_1 l_1 \ddot{x} \cos \theta_1 + m_1 l_1 \dot{x} \dot{\theta}_1 \sin \theta_1 + m_1 l_1^2 \ddot{\theta}_1) - (m_1 l_1 \dot{x} \dot{\theta}_1 \sin \theta_1 - m_1 g l_1 \sin \theta_1) = 0$$

Simplifying (canceling $\dot{x} \dot{\theta}$ terms and dividing by $m_1 l_1$):

$$-\ddot{x} \cos \theta_1 + l_1 \ddot{\theta}_1 + g \sin \theta_1 = 0 \quad (7)$$

Equation for θ_2 Following the exact same procedure as θ_1 , we obtain:

$$-\ddot{x} \cos \theta_2 + l_2 \ddot{\theta}_2 + g \sin \theta_2 = 0 \quad (8)$$

1.5 Final Dynamic Non Linear Equations

From equations (7) and (8), we can isolate the angular accelerations:

$$\ddot{\theta}_1 = \frac{1}{l_1}(\ddot{x} \cos \theta_1 - g \sin \theta_1) \quad (9)$$

$$\ddot{\theta}_2 = \frac{1}{l_2}(\ddot{x} \cos \theta_2 - g \sin \theta_2) \quad (10)$$

From equations (5), we can isolate the linear accelerations:

$$\ddot{x} = \frac{1}{(m_1 + m_2 + M)} \left[-m_1 l_1 \dot{\theta}_1^2 \sin \theta_1 + m_1 l_1 \ddot{\theta}_1 \cos \theta_1 - m_2 l_2 \dot{\theta}_2^2 \sin \theta_2 + m_2 l_2 \ddot{\theta}_2 \cos \theta_2 + F \right] \quad (11)$$

Substituting Equations (9) and (10) into Equation (11):

$$(m_1 + m_2 + M)\ddot{x} = \left[-m_1 l_1 \dot{\theta}_1^2 \sin \theta_1 + m_1 l_1 \cos \theta_1 \left(\frac{\ddot{x} \cos \theta_1 - g \sin \theta_1}{l_1} \right) - m_2 l_2 \dot{\theta}_2^2 \sin \theta_2 + m_2 l_2 \cos \theta_2 \left(\frac{\ddot{x} \cos \theta_2 - g \sin \theta_2}{l_2} \right) + F \right]$$

Simplifying the final result:

$$(m_1 + m_2 + M)\ddot{x} = \left[-m_1 l_1 \dot{\theta}_1^2 \sin \theta_1 + m_1 \ddot{x} \cos^2 \theta_1 - m_1 g \sin \theta_1 \cos \theta_1 - m_2 l_2 \dot{\theta}_2^2 \sin \theta_2 + m_2 \ddot{x} \cos^2 \theta_2 - m_2 g \sin \theta_2 \cos \theta_2 + F \right]$$

Grouping the $\ddot{x}$ terms:

$$\left[(m_1 + m_2 + M) - m_1 \cos^2 \theta_1 - m_2 \cos^2 \theta_2 \right] \ddot{x} = -m_1 l_1 \dot{\theta}_1^2 \sin \theta_1 - m_1 g \sin \theta_1 \cos \theta_1 - m_2 l_2 \dot{\theta}_2^2 \sin \theta_2 - m_2 g \sin \theta_2 \cos \theta_2 + F$$

Using the identity $(1 - \cos^2 \theta) = \sin^2 \theta$, the Left Hand Side (LHS) becomes:

$$(m_1 \sin^2 \theta_1 + m_2 \sin^2 \theta_2 + M)\ddot{x}$$

Thus, the equation for $\ddot{x}$ is:

$$\ddot{x} = \frac{-m_1 l_1 \dot{\theta}_1^2 \sin \theta_1 - m_1 g \sin \theta_1 \cos \theta_1 - m_2 l_2 \dot{\theta}_2^2 \sin \theta_2 - m_2 g \sin \theta_2 \cos \theta_2 + F}{(m_1 \sin^2 \theta_1 + m_2 \sin^2 \theta_2 + M)} \quad (13)$$

2 Obtain the linearized system around the equilibrium point specified by $\mathbf{x} = 0$ and $\theta_1 = \theta_2 = 0$. Write the state-space representation of the linearized system.

To find the equilibrium points, we set all time derivatives to zero ($\dot{x} = \ddot{x} = \dot{\theta} = \ddot{\theta} = 0$).

From the angular acceleration equations derived in Part 1:

$$\ddot{\theta}_1 = \frac{1}{l_1}(\ddot{x} \cos \theta_1 - g \sin \theta_1) \implies 0 = 0 - g \sin \theta_1$$

Since $g \neq 0$, we must have $\sin \theta_1 = 0$, which implies $\theta_1 = 0, \pi, 2\pi \dots$. Similarly, $\theta_2 = 0, \pi, 2\pi \dots$. Substituting these into Equation (13) with zero velocity:

$$0 = 0 + F \implies F = 0$$

Thus, the equilibrium state is defined as:

$$x \in \mathbb{R}, \quad \theta_1 = n\pi, \quad \theta_2 = n\pi, \quad \dot{x} = 0, \quad \dot{\theta}_1 = 0, \quad \dot{\theta}_2 = 0, \quad F = 0$$

2.1 Linearization

We linearize the system around the stable equilibrium point:

$$\mathbf{x}_{eq} = [0, 0, 0, 0, 0, 0]^T$$

We apply the small angle approximations:

- $\sin \theta \approx \theta$
- $\cos \theta \approx 1$
- $\dot{\theta}^2 \approx 0$ (Higher order terms negligible)
- Denominator $(m_1 \sin^2 \theta_1 + m_2 \sin^2 \theta_2 + M) \approx M$

Linearized Equations of Motion

Equation (13) simplifies to:

$$\ddot{x} = \frac{1}{M}(-m_1 g \theta_1 - m_2 g \theta_2 + F) \quad (12)$$

Substituting $\ddot{x}$ into the angular equations $\ddot{\theta}_1 = \frac{1}{l_1}(\ddot{x} - g\theta_1)$:

$$\begin{aligned} \ddot{\theta}_1 &= \frac{1}{l_1} \left[\frac{1}{M}(-m_1 g \theta_1 - m_2 g \theta_2 + F) - g\theta_1 \right] \\ &= \frac{1}{l_1 M} [-m_1 g \theta_1 - m_2 g \theta_2 + F - M g \theta_1] \\ \ddot{\theta}_1 &= \frac{1}{l_1 M} [-(m_1 + M)g\theta_1 - m_2 g \theta_2 + F] \end{aligned} \quad (13)$$

Similarly for θ_2 :

$$\ddot{\theta}_2 = \frac{1}{l_2 M} [-m_1 g \theta_1 - (m_2 + M)g\theta_2 + F] \quad (14)$$

2.2 State-Space Representation

Let the state vector be $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5, x_6]^T = [x, \theta_1, \theta_2, \dot{x}, \dot{\theta}_1, \dot{\theta}_2]^T$.

The state-space form is $\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$:

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & -\frac{m_1 g}{M} & -\frac{m_2 g}{M} & 0 & 0 & 0 \\ 0 & -\frac{(m_1 + M)g}{l_1 M} & -\frac{m_2 g}{l_1 M} & 0 & 0 & 0 \\ 0 & -\frac{m_1 g}{l_2 M} & -\frac{(m_2 + M)g}{l_2 M} & 0 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{M} \\ \frac{1}{l_1 M} \\ \frac{1}{l_2 M} \end{bmatrix} \quad (15)$$

3 Obtain conditions on M, m_1, m_2, l_1, l_2 for which the linearized system is controllable.

For the system to be controllable, the Controllability Matrix $\mathcal{C}$ must have full rank (Rank = 6).

$$\mathcal{C} = [B \quad AB \quad A^2B \quad A^3B \quad A^4B \quad A^5B] \quad (16)$$

3.1 Calculating Columns of $\mathcal{C}$

Vector B :

$$B = \left[0, 0, 0, \frac{1}{M}, \frac{1}{l_1 M}, \frac{1}{l_2 M} \right]^T$$

Vector AB : Note that for the structure of A , AB results in a vector with non-zero elements in rows 1, 2, and 3 corresponding to velocities (which are zero in B).

$$AB = \left[\frac{1}{M}, \frac{1}{l_1 M}, \frac{1}{l_2 M}, 0, 0, 0 \right]^T$$

Vector A^2B : Calculating $A \cdot (AB)$:

$$A^2B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -\frac{m_1 g}{l_1 M^2} - \frac{m_2 g}{l_2 M^2} \\ -\frac{(m_1 + M)g}{l_1^2 M^2} - \frac{m_2 g}{l_1 l_2 M^2} \\ -\frac{m_1 g}{l_1 l_2 M^2} - \frac{(m_2 + M)g}{l_2^2 M^2} \end{bmatrix}$$

Vector A^3B : Calculating $A \cdot (A^2B)$:

$$A^3B = \begin{bmatrix} -\frac{m_1 g}{l_1 M^2} - \frac{m_2 g}{l_2 M^2} \\ -\frac{(m_1 + M)g}{l_1^2 M^2} - \frac{m_2 g}{l_1 l_2 M^2} \\ -\frac{m_1 g}{l_1 l_2 M^2} - \frac{(m_2 + M)g}{l_2^2 M^2} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

3.2 Rank Condition

Examining the structure of matrix $\mathcal{C}$ using rows 2, 3, 5, and 6: We can see that the column vectors are showing pattern. Hence we can guarantee the A^4B and A^5B will also be similar. From looking at the column vectors we can see that if $l_1 = l_2$ the rows will become identical resulting in $\det(\mathcal{C})=0$.

- Row 2 corresponds to θ_1 , Row 3 corresponds to θ_2 .
- Row 5 corresponds to $\dot{\theta}_1$, Row 6 corresponds to $\dot{\theta}_2$.

Looking at the columns generated:

- If $l_1 = l_2$, then Row 2 is scalar multiple of Row 3.

- If $l_1 = l_2$, then Row 5 is scalar multiple of Row 6.

Specifically, if $l_1 = l_2$, the pendulum becomes identical (parallel), which might be leading the system into oscillations.

Conclusion:

- If $l_1 = l_2$: The system is **uncontrollable** (Rank < 6).
- If $l_1 \neq l_2$: The system is **controllable**.

The masses m_1 and m_2 do not affect the controllability condition; they can be equal or unequal ($m_1 = m_2$ is allowed). The critical geometric parameter is the pendulum length.

$m_1, m_2, M, l_1, l_2 > 0$ We can validate this with MATLAB using $C_o = \text{ctrb}(A, B)$ function.

Also, there is a file for calculating the determinant of symbolic representation of Controllability matrix namely **Controllability.m**. On Running these we get the below result as shown in diagram.

The determinant of Controllability Matrix (Co) is:
 $(g^6*(l_1 - l_2)^2)/(M^6*l_1^6*l_2^6)$

The Symbolic Controllability Matrix (Co) is:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{M} & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Figure 2: MATLAB code result for "Controllability.m" file

where

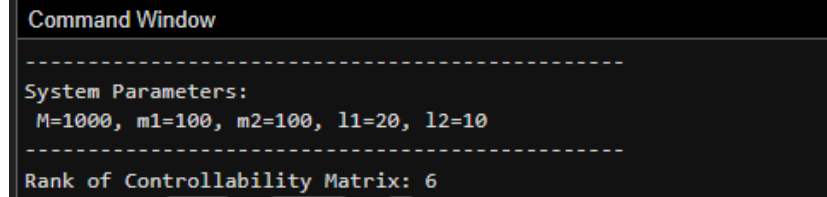
$$\begin{aligned}
\#1 &= - \frac{g(M l_2 + l_1 m_2 + l_2 m_1)}{M^2 l_1 l_2} \\
\#2 &= - \frac{g(M l_1 + l_1 m_2 + l_2 m_1)}{M^2 l_1 l_2} \\
\#3 &= - \frac{g(l_1 m_2 + l_2 m_1)}{M^2 l_1 l_2} \\
\#4 &= - \frac{g(M^2 l_2 + M l_1 m_2 + M l_1 l_2 m_2 + 2 M l_2 m_1 + \#8 + 2 l_1 l_2 m_1 m_2 + \#7)}{M^3 l_1 l_2} \\
\#5 &= - \frac{g(M^2 l_1 + 2 M l_1 m_2 + M l_1 l_2 m_1 + M l_2 m_1 + \#8 + 2 l_1 l_2 m_1 m_2 + \#7)}{M^3 l_1 l_2} \\
\#6 &= - \frac{g(\#8 + M l_1 m_2 + 2 l_1 l_2 m_1 m_2 + \#7 + M l_2 m_1)}{M^3 l_1 l_2} \\
\#7 &= l_2^2 m_1 \\
\#8 &= l_1^2 m_2
\end{aligned}$$

Figure 3: MATLAB code result for "Controllability.m" file

We can see that the determinant can be zero only if $l_1 = l_2$. Hence we have validated the controllability condition.

4 Choose $M=1000\text{Kg}$, $m_1 = m_2 = 100\text{Kg}$, $l_1=20\text{m}$ and $l_2=10\text{m}$. Check that the system is controllable and obtain an LQR controller

There is MATLAB file attached with the report namely `Controllability_with_values.m` which is fed with the values of mass and lengths. We get a rank 6 in the code and hence the system is controllable for the given set of mass and length values.



```

Command Window

-----
System Parameters:
M=1000, m1=100, m2=100, l1=20, l2=10
-----
Rank of Controllability Matrix: 6

```

Figure 4: MATLAB code result for `Controllability_with_values.m`

With these values lets design LQR controller. we have:

$$\begin{aligned}
 \text{System Dynamics: } \dot{x} &= Ax + Bu \\
 \text{Cost Function: } J &= \int_0^\infty (x^T Q x + u^T R u) dt \\
 \text{Riccati Equation: } A^T P + PA - PBR^{-1}B^T P + Q &= 0 \\
 \text{Optimal Gain: } K &= R^{-1}B^T P \\
 \text{Control Law: } u &= -Kx
 \end{aligned}$$

We will utilize the LQR function in Matlab to calculate the value for P and K. We will fix the Q matrix and R matrix as follows.

$q_x = 1$	Penalty on cart position
$q_{\theta_1} = 100$	Penalty on Pendulum 1 angle
$q_{\theta_2} = 100$	Penalty on Pendulum 2 angle
$q_{\dot{x}} = 1$	Penalty on cart velocity
$q_{\dot{\theta}_1} = 10$	Penalty on angular velocity 1
$q_{\dot{\theta}_2} = 10$	Penalty on angular velocity 2
$Q = \text{diag}(q_x, q_{\theta_1}, q_{\theta_2}, q_{\dot{x}}, q_{\dot{\theta}_1}, q_{\dot{\theta}_2})$	
$R = 1$	Penalty on Input

Thus we have a Q matrix and scalar value set for R. Now we will run the code to check the obtained P matrix, K matrix and Eigen Values. We will change the values of Q matrix and R matrix to obtain values that will indicate a stable syste. Contrsaints that we will consider:

- P should be Positive definite
- Eigen values should be in the left half

With Q and R matrix set to 1 the result was as shown below:

```

Command Window

The selected Q Matrix:
    1    0    0    0    0    0
    0    1    0    0    0    0
    0    0    1    0    0    0
    0    0    0    1    0    0
    0    0    0    0    1    0
    0    0    0    0    0    1

The selected R Matrix:
    1

LQR Gain Matrix K:
    1.0000   -1.5319   -0.7671   48.9952   -78.2683   -39.1881

LQR P Matrix:
    1.0e+05 *
    0.0005   -0.0008   -0.0004    0.0120   -0.0200   -0.0100
   -0.0008    0.4020   -0.0005   -0.0184    0.0306    0.0153
   -0.0004   -0.0005    0.1947   -0.0092    0.0154    0.0077
    0.0120   -0.0184   -0.0092    0.5879   -0.9800   -0.4900
   -0.0200    0.0306    0.0154   -0.9800    2.3825    0.7816
   -0.0100    0.0153    0.0077   -0.4900    0.7816    0.5901

LQR Eigen values of A-B.K:
   -0.0001 + 0.7285i
   -0.0001 - 0.7285i
   -0.0001 + 1.0430i
   -0.0001 - 1.0430i
   -0.0204 + 0.0204i
   -0.0204 - 0.0204i

```

Figure 5: MATLAB code result for above case

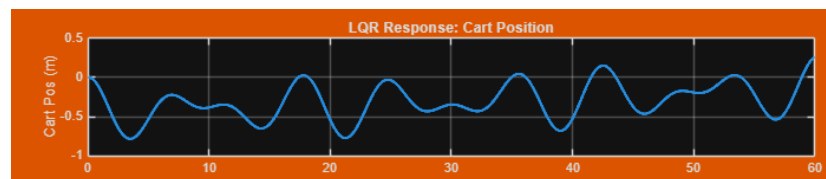


Figure 6: MATLAB code result for above case

We can see that 4 of the eigen values are very much close to the imaginary axis which means that the system will take high time to become stable. From the plots of displacement with these values it was seen that the system was not achieving stability in set time of 40 seconds.

With value of R fixed at 1 values of Q matrix was changed but the system remained oscillatory we could observe the eigen values are very much close to the imaginary axis. One of the case is shown in the figure below:

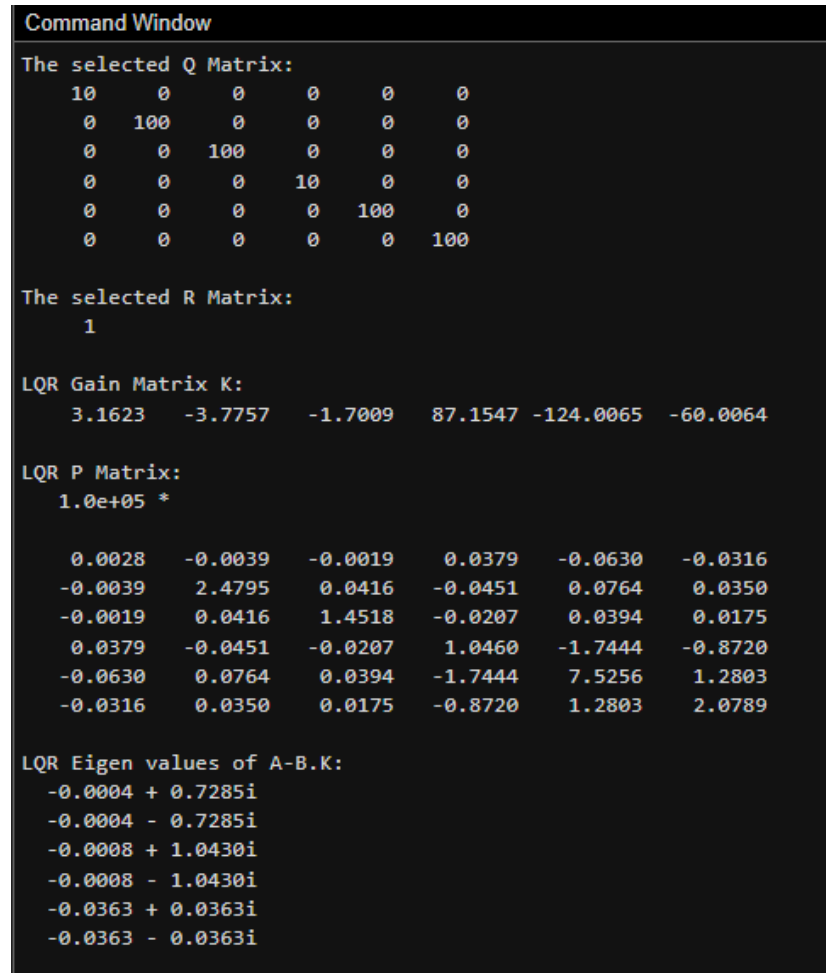


Figure 7: MATLAB result for above case

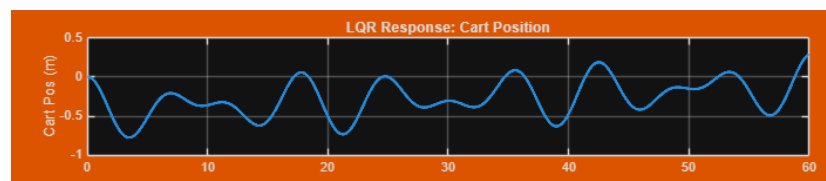


Figure 8: MATLAB result for above case

In this case also we can see that 4 of the eigen values are very much close to the imaginary axis which means that the system will take high time to become stable. From the plots of displacement with these values it was seen that the system was not achieving stability in set time of 40 seconds.

Now value of R was being varied, and the system was being observed. One of the case is shown in the figure below:

```

Command Window

The selected Q Matrix:
    1    0    0    0    0    0
    0   100    0    0    0    0
    0    0   100    0    0    0
    0    0    0    1    0    0
    0    0    0    0   10    0
    0    0    0    0    0   10

The selected R Matrix:
    1.0000e-05

LQR Gain Matrix K:
    1.0e+03 *
    0.3162    2.1169    2.4603    1.0743    1.9452    0.3209

LQR P Matrix:
    1.0e+03 *
    0.0034    0.0062    0.0010    0.0053   -0.0200   -0.0111
    0.0062    0.9076   -0.0471    0.0409    0.0139   -0.2040
    0.0010   -0.0471    0.4591    0.0145    0.2102   -0.0045
    0.0053    0.0409    0.0145    0.0172   -0.0553   -0.0366
   -0.0200    0.0139    0.2102   -0.0553    1.7276   -0.1162
   -0.0111   -0.2040   -0.0045   -0.0366   -0.1162    0.4567

LQR Eigen values of A-B.K:
   -0.0736 + 0.7235i
   -0.0736 - 0.7235i
   -0.1603 + 1.0395i
   -0.1603 - 1.0395i
   -0.3678 + 0.3532i
   -0.3678 - 0.3532i

```

Figure 9: MATLAB result for above case

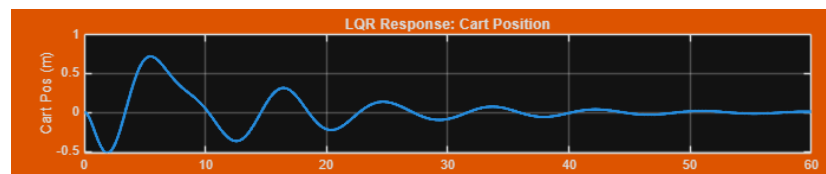


Figure 10: MATLAB result for above case

In this case we can see a significant shift in the eigen values towards the left. Also on plotting these values we could see the system going to stability around 40 seconds.

The values of R and Q was iterated over and over again to reach the stable value shown in the figure.

```

Command Window

The selected Q Matrix:
      10      0      0      0      0      0
      0    3000      0      0      0      0
      0      0    2000      0      0      0
      0      0      0      1      0      0
      0      0      0      0    100      0
      0      0      0      0      0     50

The selected R Matrix:
1.0000e-06

LQR Gain Matrix K:
1.0e+04 *
      0.3162      5.4232      2.7442      0.7826      0.1795     -3.5584

LQR P Matrix:
1.0e+04 *
      0.0025      0.0006     -0.0113      0.0030     -0.0356     -0.0092
      0.0006      1.2676     -0.3980      0.0370     -0.0048     -0.3134
     -0.0113     -0.3980      0.4005     -0.0187      0.3070      0.0608
      0.0030      0.0370     -0.0187      0.0085     -0.0869     -0.0341
     -0.0356     -0.0048      0.3070     -0.0869      2.5642     -0.4109
     -0.0092     -0.3134      0.0608     -0.0341     -0.4109      0.5113

LQR Eigen values of A-B.K:
-0.1534 + 0.8000i
-0.1534 - 0.8000i
-0.2716 + 0.5527i
-0.2716 - 0.5527i
-1.7536 + 1.7235i
-1.7536 - 1.7235i

7.4854e+16

```

Figure 11: MATLAB result for above case

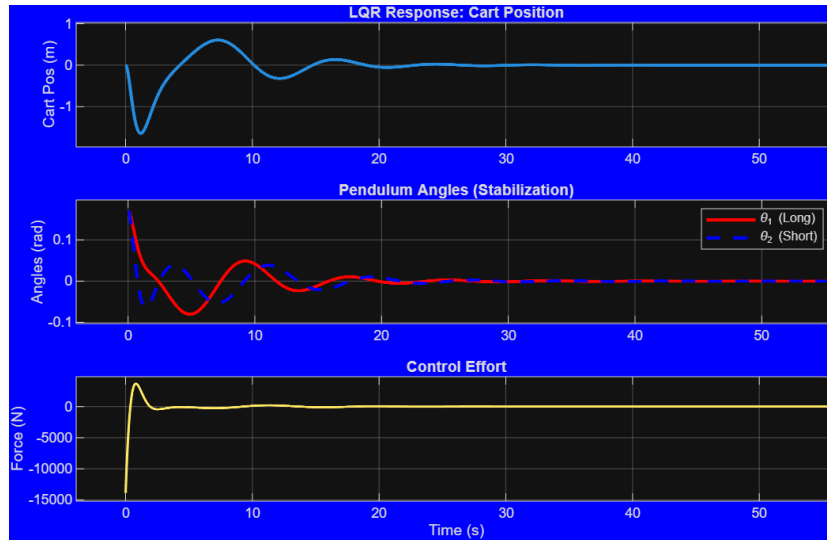


Figure 12: MATLAB result for above case

Note: Run LQR.m to obtain these results above. Change the values of Q and R accordingly.

As we can see the when the controller is applied to the linearized system for input conditions the system attains stability in 20 seconds. The gain Q Matrix values indicate we have had to penalise the the angular position and angular velocities of the pendulum heavily. Also the penalty on the input by R Matrix was very minimal. These two observations were made after obtaining the stable system. The Final Q and R marices are as follows:

$q_x = 10$	Penalty on cart position
$q_{\theta_1} = 3000$	Penalty on Pendulum 1 angle
$q_{\theta_2} = 2000$	Penalty on Pendulum 2 angle
$q_{\dot{x}} = 1$	Penalty on cart velocity
$q_{\dot{\theta}_1} = 100$	Penalty on angular velocity 1
$q_{\dot{\theta}_2} = 50$	Penalty on angular velocity 2
$R = 0.000001$	Penalty on Input

Now the linear system was compared with non linear system. ODE45 tool of MATLAB is used to compute the Nonlinear dynamics. Initial conditions was set as follows first:

$$x_0 = [0; 0.5; 0.5; 0; 0; 0];$$

We have considered 0.5 radians angles for pendulum 1 and 2.

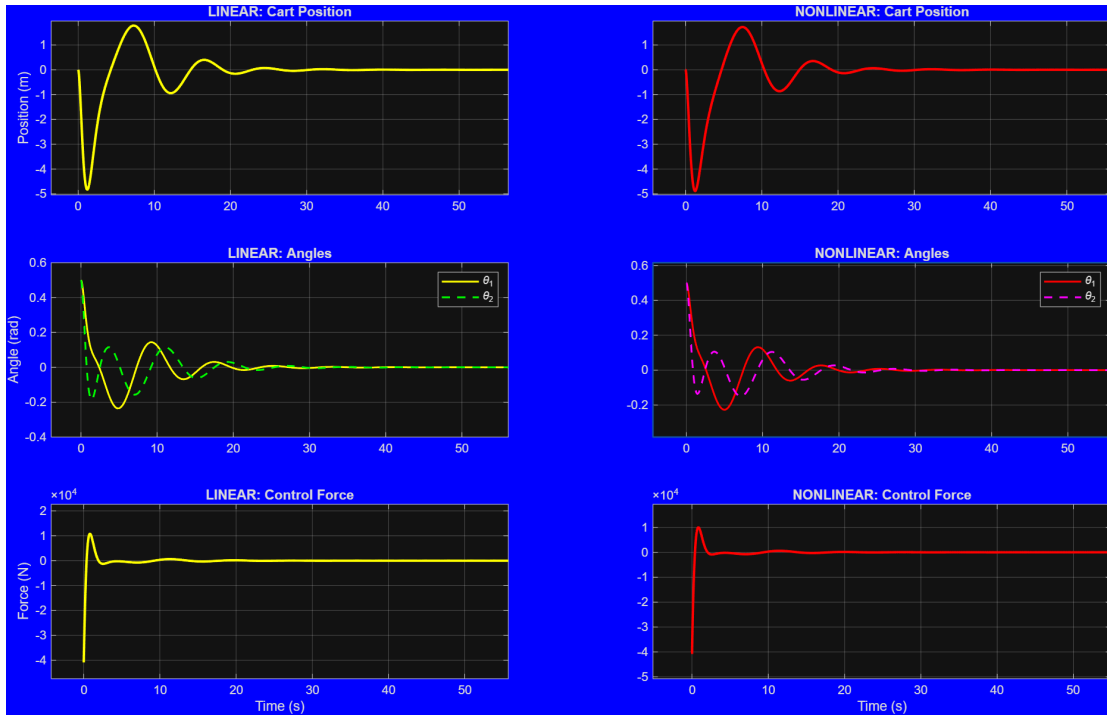


Figure 13: MATLAB result for above case

We can see that not much difference in outputs is seen for small angles.

For the next case we have considered a higher value for angle (3 radians).
 $x_0 = [0; 3; 3; 0; 0; 0];$

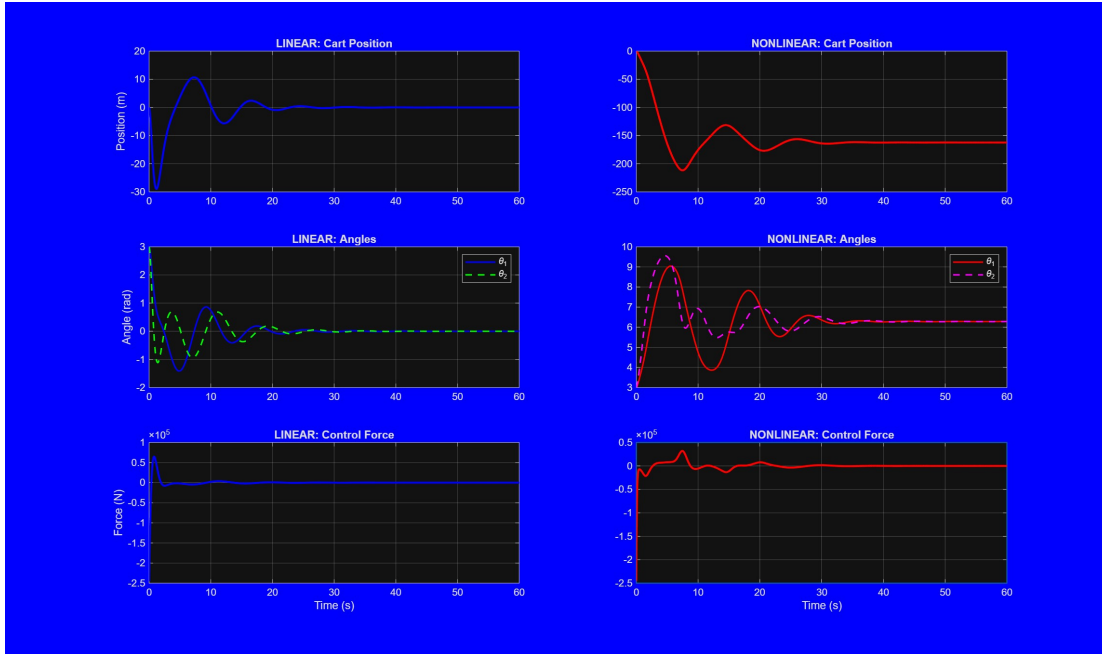


Figure 14: MATLAB result for above case

We can see that the system is going into stable condition but the position has been shifted in the case of non linear dynamics. The pendulum will die out in both case as its in hanging condition but there is a fixed error in Non Linear system. For small value of angles (the approximation we considered for linearising the system) both linear and non linear system has the same behaviour.

The lyapunov indirect method is used to certify stability (locally or globally) of the closed-loop system. We have checked the Eigen values of the closed loop system.

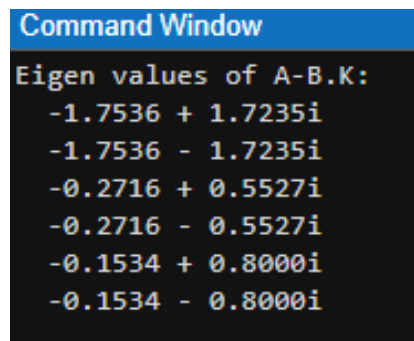


Figure 15: MATLAB result for above case

The eigenvalues of the linearized closed-loop matrix ($A - BK$) were computed. All eigenvalues were found to have strictly negative real parts. Therefore, according to Lyapunov's Indirect Method, the equilibrium point of the full nonlinear system is Locally Asymptotically Stable.

Note: Run the Linear_vs_Nonlinear.m file to obtain these results above. Change the values of x_0 accordingly.

5 Observability Analysis

To determine the observability of the linearized system, we examine the rank of the observability matrix $\mathcal{O}$ for the linearized state-space model we have.

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ CA^3 \\ CA^4 \\ CA^5 \end{bmatrix}$$

We evaluate the observability for the following output vector choices:

Case 1: $y(t) = [x(t)]$

The Matlab Analysis is as follows.

```

Command Window

Case:output is x(t)
The Matrix C is:      1      0      0      0      0      0

Symbolic Observability Determinant:
      6      2      2      2
      g  m1  m2  (l1 - l2)
-----
      4      2      2
      M  l1  l2

```

Figure 16: MATLAB result for above case

Here we can see the determinant will be zero if $l_1 = l_2$ and the state space will not be observable. So for this case $l_1 \neq l_2$ will be the condition for observability.

Case 2: $y(t) = [\theta_1(t), \theta_2(t)]$

The Matlab Analysis is as follows.

```

Command Window

Case:output is theta_1(t), theta_2(t)
The Matrix C is:
      0      1      0      0      0      0
      0      0      1      0      0      0

The Observability Matrix is:
[0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
[0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0]
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1]
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
rank_symbolic =
4

```

Figure 17: MATLAB result for above case

The symbolic rank of the matrix itself is less than 6 hence this case is also not observable.

Case 3: $y(t) = [x(t), \theta_2(t)]$

The Matlab Analysis is as follows.

```
Case:output is x(t), theta_2(t)
The Matrix C is:
    1    0    0    0    0    0
    0    0    1    0    0    0

The Observability Matrix is:
[1,                                     0,                                     0, 0,
[0,                                     0,                                     1, 0,
[0,                                     0,                                     0, 1,
[0,                                     0,                                     0, 0,
[0,                                     -(g*m1)/M,                             -(g*m2)/M, 0,
[0,                                     -(g*m1)/(M*l2),                         -(g*(M + m2))/(M*l2), 0,
[0,                                     0,                                     0, 0,
[0,                                     0,                                     0, 0,
[0,                                     (g^2*m1*(M + m1))/(M^2*l1) + (g^2*m1*m2)/(M^2*l2), (g^2*m2*(M + m2))/(M^2*l2) + (g^2*m1*m2)/(M^2*l1), 0,
[0, (g^2*m1*(M + m2))/(M^2*l2^2) + (g^2*m1*(M + m1))/(M^2*l1*l2), (g^2*(M + m2)^2)/(M^2*l2^2) + (g^2*m1*m2)/(M^2*l1*l2), 0,
[0,                                     0,                                     0, 0,
[0,                                     0,                                     0, 0,
```

Figure 18: MATLAB result for above case

```
0,                                     0,                                     0]
0,                                     0,                                     0]
1,                                     0,                                     0]
0, | 0, 1]
0, 0, 0]
0, 0, 0]
0,                                     -(g*m1)/M,                             -(g*m2)/M]
0,                                     -(g*m1)/(M*l2),                         -(g*(M + m2))/(M*l2)]
0,                                     0,                                     0]
0,                                     0,                                     0]
0,                                     (g^2*m1*(M + m1))/(M^2*l1) + (g^2*m1*m2)/(M^2*l2), (g^2*m2*(M + m2))/(M^2*l2) + (g^2*m1*m2)/(M^2*l1)]
0, (g^2*m1*(M + m2))/(M^2*l2^2) + (g^2*m1*(M + m1))/(M^2*l1*l2), (g^2*(M + m2)^2)/(M^2*l2^2) + (g^2*m1*m2)/(M^2*l1*l2)]
```

Figure 19: MATLAB result for above case

This is the complete Observability matrix showed in 2 parts due to width constraints. Lets look at the Matrix to see when the rank condition fails. We can see that if we have no mass m1 or lets say if m1=0 then the columns 2 and column 5 will become zero vector resulting in failure of Observability. Lets check the observability of this matrix by substituting the values.

```

Command Window
Case:output is x(t), theta_2(t)
The Matrix C is:
    1    0    0    0    0    0
    0    0    1    0    0    0

The Mass of the Cart is: 1000 kg
The Mass of pendulum 1 is: 100 kg
The Mass of pendulum 2 is: 100 kg
The Length of pendulum 1 is: 20 m
The Length of pendulum 2 is: 10 m
The Numeric Rank is: 6

```

Figure 20: MATLAB result for above case

```

Command Window
Case:output is x(t), theta_2(t)
The Matrix C is:
    1    0    0    0    0    0
    0    0    1    0    0    0

The Mass of the Cart is: 1000 kg
The Mass of pendulum 1 is: 0 kg
The Mass of pendulum 2 is: 100 kg
The Length of pendulum 1 is: 20 m
The Length of pendulum 2 is: 10 m
The Numeric Rank is: 4

```

Figure 21: MATLAB result for above case

```

Command Window
Case:output is x(t), theta_2(t)
The Matrix C is:
    1    0    0    0    0    0
    0    0    1    0    0    0

The Mass of the Cart is: 1000 kg
The Mass of pendulum 1 is: 100 kg
The Mass of pendulum 2 is: 100 kg
The Length of pendulum 1 is: 10 m
The Length of pendulum 2 is: 10 m
The Numeric Rank is: 6

```

Figure 22: MATLAB result for above case

So for this case $m_1 \neq 0$ will be the condition for observability.


```

Command Window
Case:output is x(t),theta_1(t) theta_2(t)
The Matrix C is:
    1    0    0    0    0    0
    0    1    0    0    0    0
    0    0    1    0    0    0

The Mass of the Cart is: 1000 kg
The Mass of pendulum 1 is: 0 kg
The Mass of pendulum 2 is: 100 kg
The Length of pendulum 1 is: 20 m
The Length of pendulum 2 is: 10 m
The Observability Matrix is:
The Numeric Rank is: 6

```

Figure 25: MATLAB result for above case

```

Command Window
Case:output is x(t),theta_1(t) theta_2(t)
The Matrix C is:
    1    0    0    0    0    0
    0    1    0    0    0    0
    0    0    1    0    0    0

The Mass of the Cart is: 1000 kg
The Mass of pendulum 1 is: 100 kg
The Mass of pendulum 2 is: 100 kg
The Length of pendulum 1 is: 10 m
The Length of pendulum 2 is: 10 m
The Observability Matrix is:
The Numeric Rank is: 6

```

Figure 26: MATLAB result for above case

```

Command Window
Case:output is x(t),theta_1(t) theta_2(t)
The Matrix C is:
    1    0    0    0    0    0
    0    1    0    0    0    0
    0    0    1    0    0    0

The Mass of the Cart is: 1000 kg
The Mass of pendulum 1 is: 100 kg
The Mass of pendulum 2 is: 0 kg
The Length of pendulum 1 is: 10 m
The Length of pendulum 2 is: 10 m
The Observability Matrix is:
The Numeric Rank is: 6

```

Figure 27: MATLAB result for above case

All of these cases shows that the state space is observable for this case without any constraints. Obviously $l_1 \neq 0$ and $l_2 \neq 0$ for all these cases.

Note: Run the Observability.m file to obtain these results above. Change the C matrix accordingly to get the results above.

6 Luenberger Observer Design and Simulation

Based on the observability analysis in Part E, we identified three observable configurations:

- Case 1: $\mathbf{y} = [x]$
- Case 3: $\mathbf{y} = [x, \theta_2]$
- Case 4: $\mathbf{y} = [x, \theta_1, \theta_2]$

For each case, we designed a Luenberger Observer defined by:

$$\dot{\hat{\mathbf{x}}} = A\hat{\mathbf{x}} + B\mathbf{u} + L(\mathbf{y} - C\hat{\mathbf{x}}) \quad (17)$$

6.1 Observer Gain Selection

To ensure the observer estimates converge quickly to the true states, we employed the pole placement method. The observer poles were placed at $[-2, -3, -4, -5, -6, -7]$, which are significantly faster than the open-loop dynamics of the cart. This ensures the transient estimation error decays within approximately 2-3 seconds.

6.2 Simulation Results

The observers were tested against the **Nonlinear System** dynamics.

- **Input:** A step input force $u = 100N$.
- **Initial Conditions:** The true system started with an initial sway ($\theta_1 = 0.1$ rad, $\theta_2 = -0.1$ rad), while the observer started at zero state.

Case 1: Measuring only Cart Position (x)

This is the most critical case as it uses the least amount of information. As shown in Figure 29, the observer successfully reconstructs the unmeasured angles θ_1 and θ_2 . The estimation error (green and magenta lines) converges to zero rapidly.

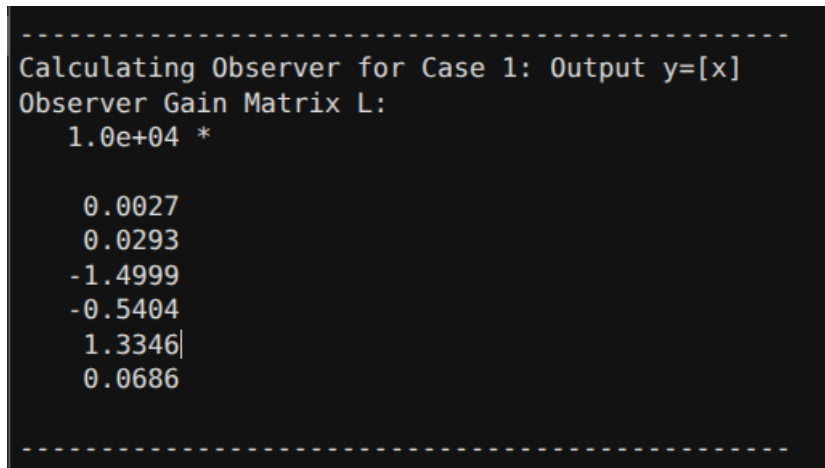


Figure 28: Observer Gain Matrix for case 1

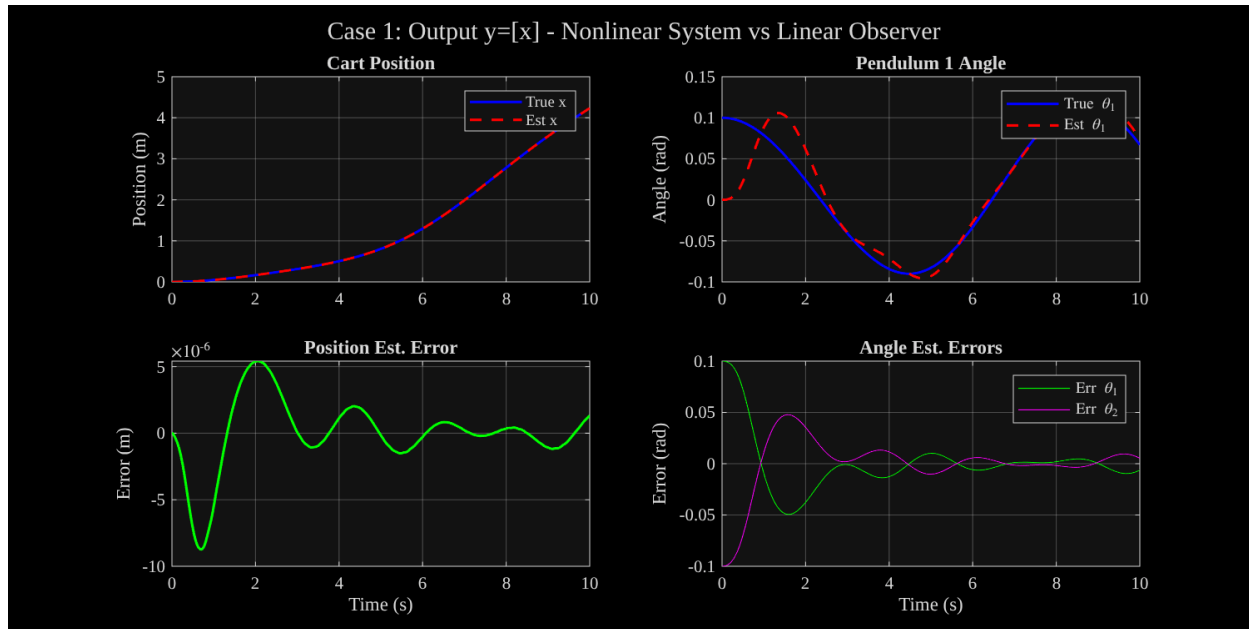


Figure 29: Observer performance for Case 1 ($y = x$). Note that despite only measuring position, the observer correctly estimates the pendulum angles.

Case 3: Measuring Cart Position and θ_2

Adding the sensor for θ_2 improves the estimation of that specific state, but the overall convergence speed is primarily dictated by the observer eigenstructure (poles) we selected.

```
-----
Calculating Observer for Case 3: Output  $y=[x, \theta_2]$ 
Observer Gain Matrix L:
  17.2418   -1.9388
 103.7782  -26.1469
-256.4987  101.6392
-174.6426  110.3637
   0.6195    9.7582
   7.4369   20.1529
-----
```

Figure 30: Observer Gain Matrix for case 3

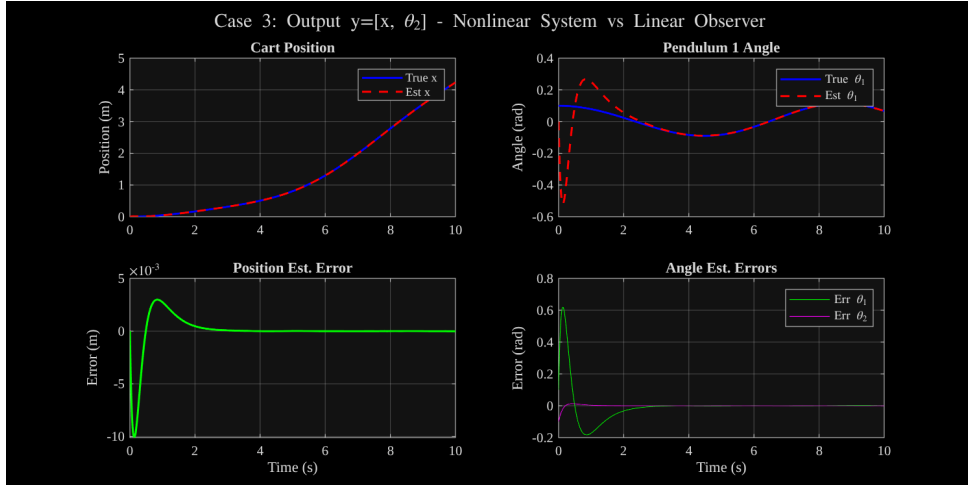


Figure 31: Observer performance for Case 3 ($y = [x, \theta_2]$).

Case 4: Measuring All States

This represents the full-state feedback scenario (or fully instrumented setup). The observer convergence is trivial here but serves as a baseline validation.

```

Calculating Observer for Case 4: Output y=[x, \theta_1, \theta_2]
Observer Gain Matrix L:
10.4331  -0.8011   0.0000
26.3638  -5.3594  -0.9809
-0.8468  11.5669  -0.0000
-4.6605  32.0963  -0.0492
0.0000   -0.0000   5.0000
0.0000   -0.0981   4.9209

```

Figure 32: Observer Gain Matrix for case 4

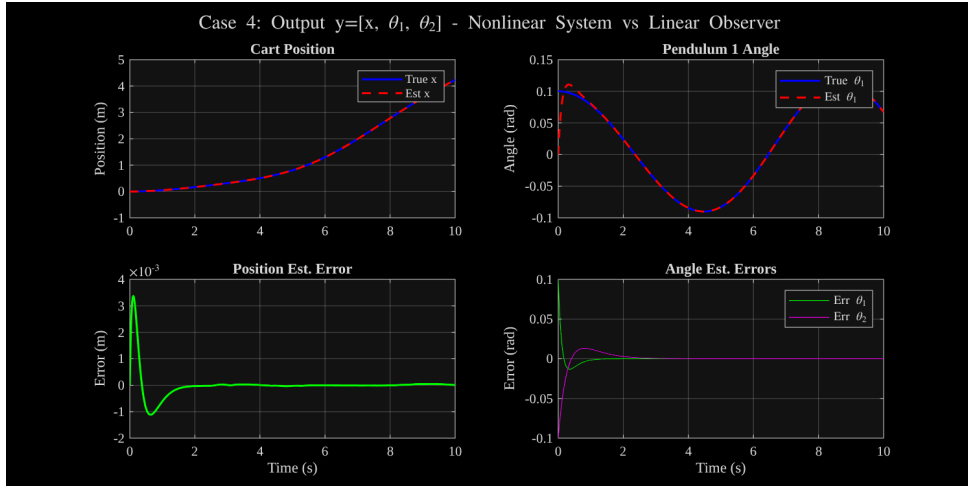


Figure 33: Observer performance for Case 4 ($y = [x, \theta_1, \theta_2]$).

Conclusion: The simulations confirm that a linear Luenberger observer is robust enough to estimate the states of the full nonlinear system, even for the minimal sensor configuration (Case 1). The estimation error converges to zero in all observable cases, validating the observability rank check performed in Part E.

7 Output Feedback Controller (LQG)

In this final component, we design an output feedback controller using the "smallest" observable output vector, $\mathbf{y}(t) = [x(t)]$ (Case 1). This configuration is the most cost-effective as it requires only a single position sensor on the cart.

$$\dot{\hat{x}}(t) = A\hat{x}(t) + Bu(t) + L(y(t) - C\hat{x}(t))$$

$$u(t) = -K\hat{x}(t)$$

K is the optimal feedback gain derived from the Algebraic Riccati Equation (ARE). The Gain K was calculated using the LQR function in MATLAB.

L is Luenberger Observer Matrix and was calculated using Kalman Estimator.

The resultant closed loop system was applied to a initial position $x_0 = [8; 0.05; -0.05; 0; 0; 0]$;

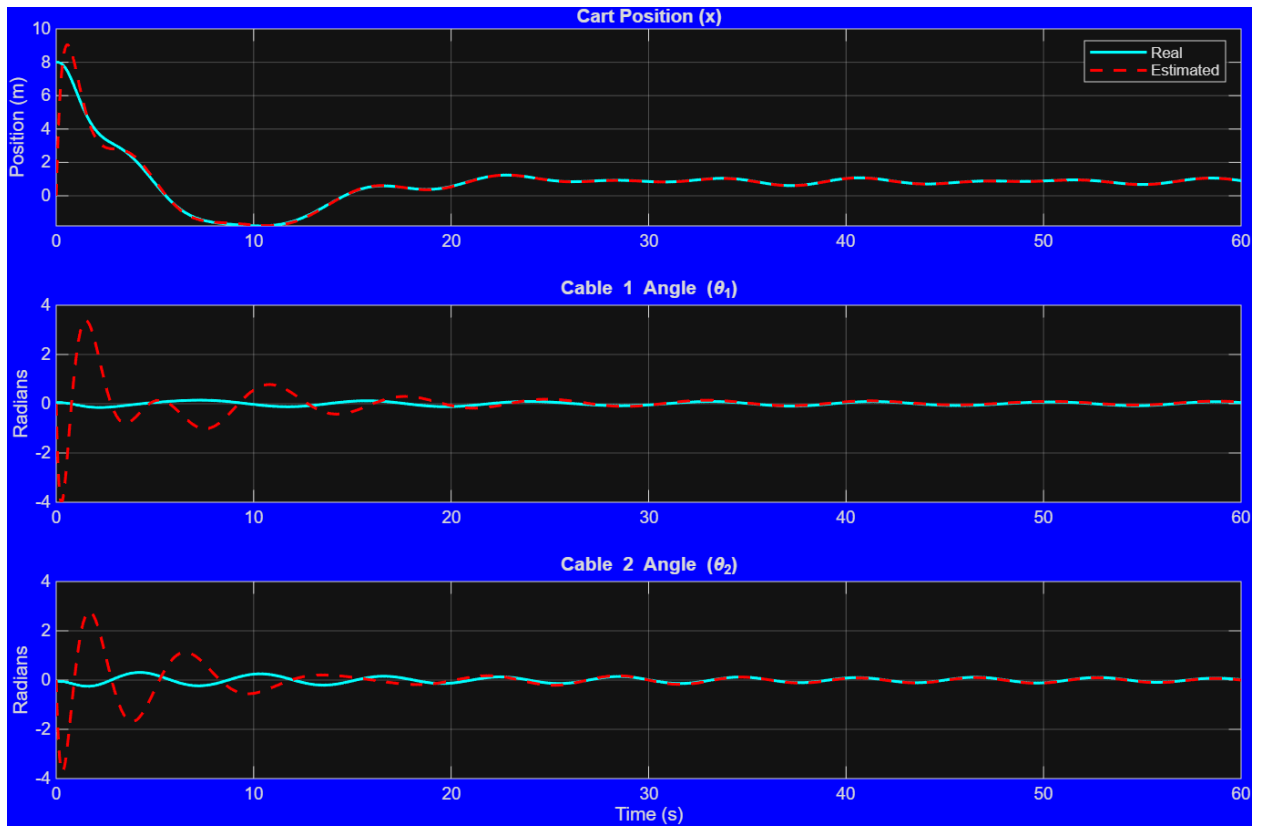


Figure 34: LQG controller with feedback and No force

We can see that the LQG controller is driving the system to zero but there exists a fixed error as we go on increasing the position far away from the system. Lets simulate the same system with fixed force input as well.

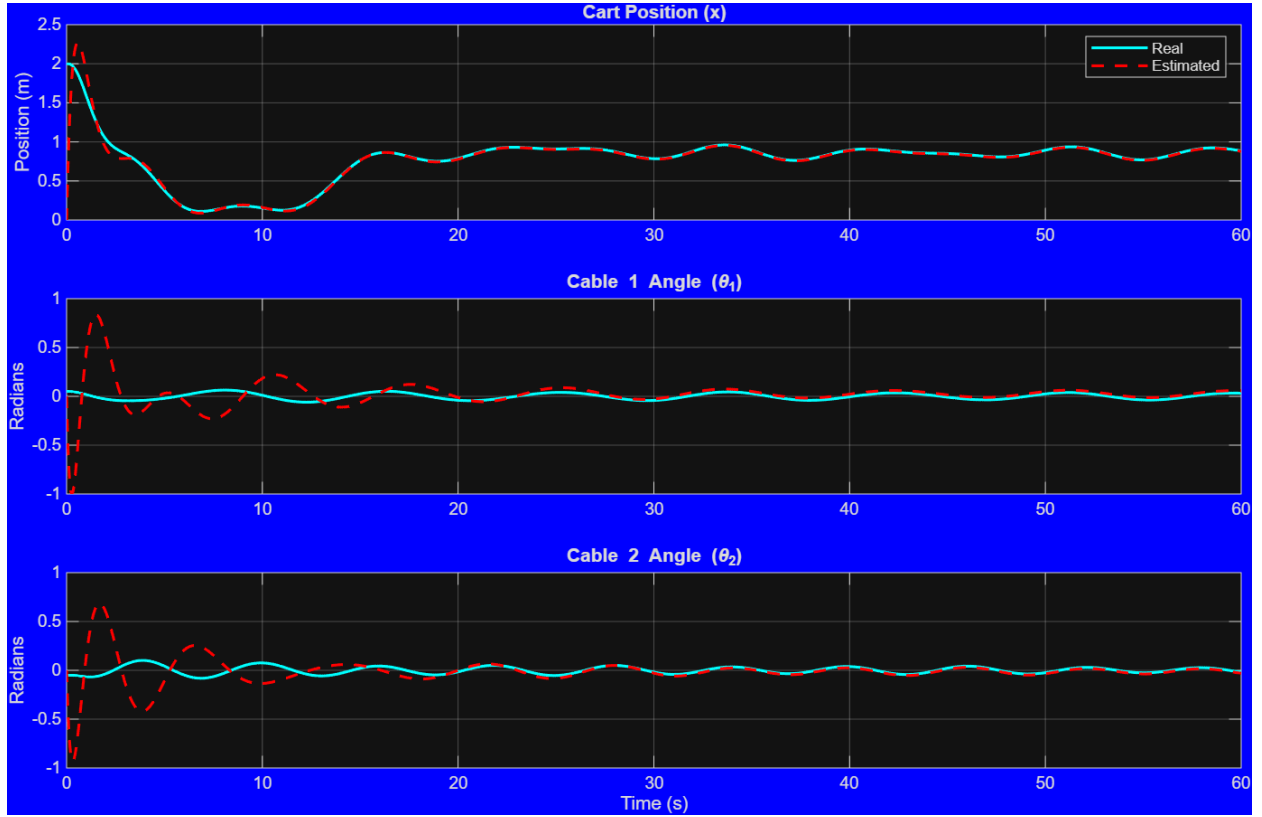


Figure 35: LQG controller with feedback and force 50Newton

We can see that Standard LQG controllers regulate states to zero but there is fixed steady state error in both the cases. In case 1 if we take the initial state as closer to the zero position the steady state error reduces(with no Force applied). But on applying Force to the same system there arises steady state error. To overcome this we need to introduce Integral component. This will track the reference along with disturbance rejection.

7.1 Controller Design: Augmenting with Integral Action

To achieve zero steady-state error, we cannot rely on proportional feedback alone. We must introduce an Integral Action to the controller. We define a new state, x_I , which is the integral of the tracking error:

$$\dot{x}_I = y(t) - x_{ref} = C\mathbf{x}(t) - x_{ref} \quad (18)$$

We augment the linearized state-space model to include this new integrator state. The augmented system dynamics become:

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{x}_I \end{bmatrix} = \underbrace{\begin{bmatrix} A & 0_{6 \times 1} \\ C & 0 \end{bmatrix}}_{A_{aug}} \begin{bmatrix} \mathbf{x} \\ x_I \end{bmatrix} + \underbrace{\begin{bmatrix} B \\ 0 \end{bmatrix}}_{B_{aug}} u + \begin{bmatrix} 0_{6 \times 1} \\ -1 \end{bmatrix} x_{ref} \quad (19)$$

7.2 Gain Calculation

We apply the LQR method to this augmented system (A_{aug}, B_{aug}) to find the optimal gain vector $K_{aug} = [K_x \quad K_I]$. To ensure the controller prioritizes eliminating steady-state error, we assigned a high penalty to the integral state in the Q matrix ($Q_{7,7} = 1000$).

The final control law uses the estimated states from the Observer (designed in Part 6) and the computed integral error:

$$u(t) = -K_x \hat{\mathbf{x}}(t) - K_I \int_0^t (y(\tau) - x_{ref}) d\tau \quad (20)$$

7.3 Simulation Results

The controller was simulated on the full **Nonlinear System** with the following scenario:

- **Reference:** Step command to move the cart to $x = 5$ meters.
- **Disturbance:** At $t = 20$ seconds, a constant wind force $F_{dist} = 500N$ is applied to the cart.

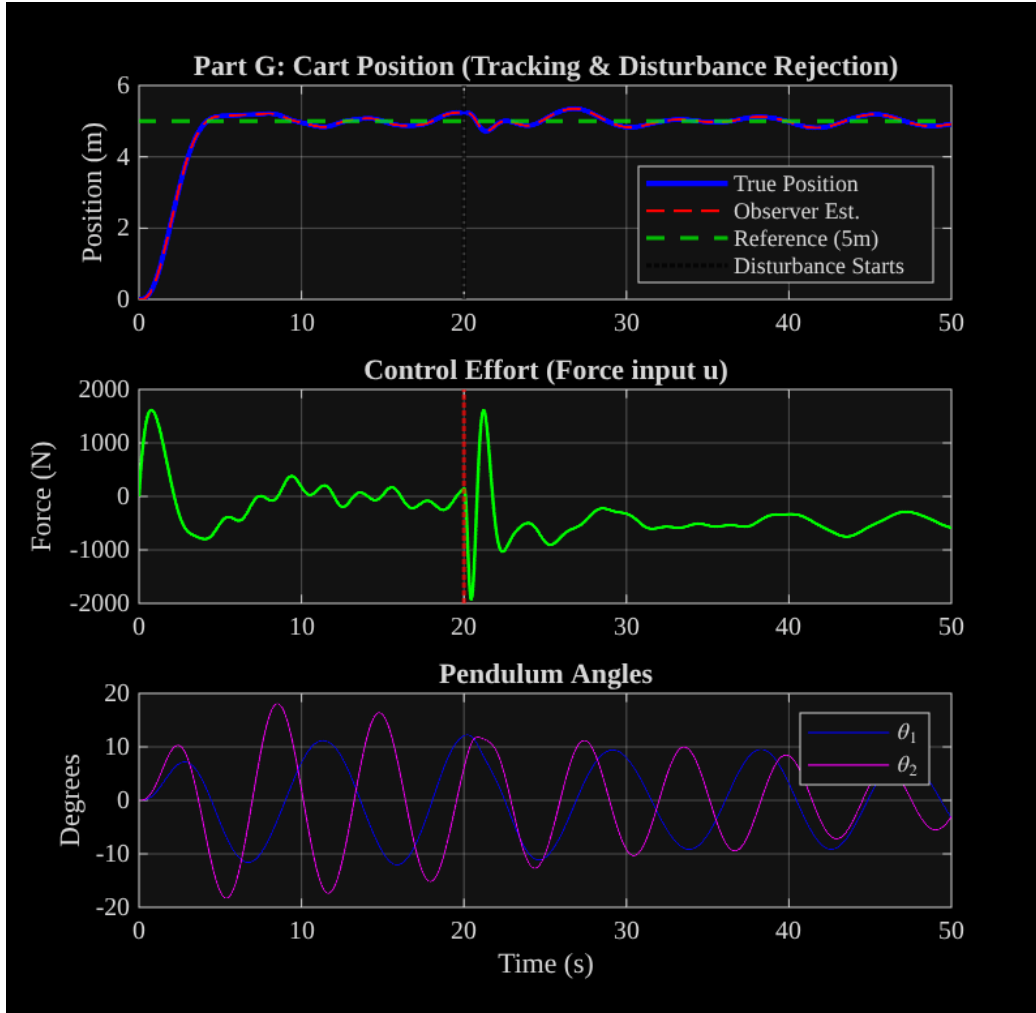


Figure 36: Performance of the LQG Controller with Integral Action. Top: Position Tracking. Middle: Control Effort. Bottom: Pendulum Angles.

Analysis of Results:

- **Tracking (0-20s):** As shown in the top subplot of Figure 36, the cart (Blue line) smoothly tracks the reference (Green dashed line) to 5m with zero steady-state error.

- **Disturbance Rejection ($t > 20s$):** At $t = 20s$, the $500N$ force hits the cart, causing a momentary dip in position. However, the **Control Effort** (Middle subplot) immediately drops to $\approx -500N$. This confirms that the integral term "wound up" to generate an exact counter-force, canceling the wind and returning the cart to exactly $5m$.

8 Conclusion

In this project, we successfully modeled and controlled an underactuated double-pendulum cart. We derived the nonlinear equations of motion and proved that the system is controllable provided the cable lengths differ ($l_1 \neq l_2$).

We demonstrated that measuring only the cart position is sufficient to observe the full system state. Finally, by implementing an LQG controller with Integral Action, we achieved robust performance, enabling the system to track targets and reject significant external disturbances despite the complex nonlinear dynamics.